

Creating VPC with NAT gateway.

1. What is a VPC

A **Virtual Private Cloud (VPC)** is a logically isolated virtual network in AWS where you can launch AWS resources such as EC2 instances, databases, and load balancers.

It allows you to **control networking components** like IP ranges, subnets, route tables, internet connectivity, and security.

2. Components Required for VPC with NAT Gateway

To build a proper VPC architecture, the following components are used:

Component	Purpose
VPC	Private virtual network in AWS
Subnet	Division of VPC network
Internet Gateway (IGW)	Allows communication between VPC and Internet
NAT Gateway	Allows private subnet instances to access internet
Route Table	Controls network traffic routing
Elastic IP	Static public IP used by NAT Gateway

3. VPC Network Design Example

Example CIDR block:

VPC CIDR : 10.0.0.0/16

Subnets:

Subnet	CIDR	Type
Public Subnet	10.0.1.0/24	Internet access
Private Subnet	10.0.2.0/24	Internal resources

4. Internet Gateway (IGW)

An **Internet Gateway** allows resources in a public subnet to communicate with the internet.

Features:

- Horizontally scalable
- Highly available
- Managed by AWS
- Attached to a VPC

Example route in public route table:

Destination	:	0.0.0.0/0
Target	:	Internet Gateway

5. NAT Gateway

A **NAT Gateway (Network Address Translation Gateway)** allows instances in a **private subnet** to access the internet **without exposing them to incoming internet traffic**.

Example:

- Private EC2 can download updates
- Install packages
- Access external APIs

But internet cannot initiate connection to private instances.

Creating an VPC

The image consists of two screenshots from the AWS Management Console. The top screenshot shows the 'Create VPC' page. The 'Resources to create' section has 'VPC only' selected. The 'Name tag - optional' field contains 'myvpc'. The 'IPv4 CIDR block' is set to '10.0.0.0/16'. The 'IPv6 CIDR block' is set to 'No IPv6 CIDR block'. The 'Tenancy' is set to 'Default'. The bottom screenshot shows the 'VPC dashboard' for the newly created VPC 'vpc-09a9894de98607334 / myvpc'. A green banner at the top indicates 'You successfully created vpc-09a9894de98607334 / myvpc'. The 'Details' section shows the VPC ID, State (Available), DNS resolution (Enabled), Main network ACL, IPv6 CIDR (Network border group), Encryption control ID, Tenancy (default), Default VPC (No), Network Address Usage metrics (Disabled), Encryption control mode, Block Public Access (Off), DHCP option set, IPv4 CIDR (10.0.0.0/16), Route 53 Resolver DNS Firewall rule groups, DNS hostnames (Disabled), Main route table, IPv6 pool, and Owner ID. The 'Resource map' section shows the VPC, Subnets (0), and Route tables (1).

Create VPC

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional [Info](#)
Creates a tag with a key of 'Name' and a value that you specify.

myvpc

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input
☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ IPAM-allocated IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block
☐ IPv6 CIDR owned by me

Tenancy [Info](#)
Default

VPC encryption control (\$) [Info](#)

VPC dashboard [AWS Global View](#)
Filter by VPC

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vpc-09a9894de98607334 / myvpc [Actions](#)

Details [Info](#)

VPC ID vpc-09a9894de98607334	State Available	Block Public Access Off	DNS hostnames Disabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0b555475aa29ed2b9	Main route table rtb-0364258cb337d9ef6
Main network ACL acl-0c49724f46528b426	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID 051741041276
Encryption control ID -	Encryption control mode -		

Resource map [Info](#) [Show all details](#)

VPC
Your AWS virtual network
myvpc

Subnets (0)
Subnets within this VPC

Route tables (1)
Route network traffic to resources
rtb-0364258cb337d9ef6

Then we have to create public and private subnet.

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1a

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24 256 IPs

Tags - optional

Key	Value - optional	
Name	public-subnet	Remove

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1a

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.2.0/24 256 IPs

Tags - optional

Key	Value - optional	
Name	private-subnet	Remove

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

VPC dashboard <

AWS Global View [Filter by VPC](#)

Virtual private cloud

- Your VPCs
- Subnets**
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

Subnets (8) Info

Last updated less than a minute ago [Actions](#) [Create subnet](#)

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	public-subnet	subnet-04059115aa5508c21	Available	vpc-09a9894de98607334 myv...	☐ Off	10.0.1.0/24
☐	-	subnet-0c69ec4b1fefaac93	Available	vpc-0be33b50250021280	☐ Off	172.31.48.0/20
☐	-	subnet-0618e0bb75502f409	Available	vpc-0be33b50250021280	☐ Off	172.31.0.0/20
☐	-	subnet-013483e4615acb32f	Available	vpc-0be33b50250021280	☐ Off	172.31.32.0/20
☐	-	subnet-07cb53f3c36994bc1	Available	vpc-0be33b50250021280	☐ Off	172.31.16.0/20
☐	private-subnet	subnet-07428dbaac21a0407	Available	vpc-09a9894de98607334 myv...	☐ Off	10.0.2.0/24
☐	-	subnet-0ca4939a51840d7cd	Available	vpc-0be33b50250021280	☐ Off	172.31.80.0/20
☐	-	subnet-07e40a724fbc9e7ed	Available	vpc-0be33b50250021280	☐ Off	172.31.64.0/20

Select a subnet

Then we have to create two Route Tables.

aws [Search] [Alt+S] United States (N. Virginia) Dhanam M (0517-4104-1226) Dhanam M

VPC > Route tables > Create route table

Create route table info

A route table specifies how packets are forwarded between the subnets within your VPC, the internet, and your VPN connection.

Route table settings

Name - optional
Create a tag with a key of 'Name' and a value that you specify.

public

VPC
The VPC to use for this route table.

vpc-09a9894de98607334 (myvpc)

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q public X Remove

Add new tag

You can add 49 more tags.

Cancel Create route table

aws [Search] [Alt+S] United States (N. Virginia) Dhanam M (0517-4104-1226) Dhanam M

VPC > Route tables > rtb-0900a7927086951c6

Route table rtb-0900a7927086951c6 | private was created successfully.

rtb-0900a7927086951c6 / private

Actions

Details info

Route table ID
rtb-0900a7927086951c6

VPC
vpc-09a9894de98607334 | myvpc

Main
No

Owner ID
051741041276

Explicit subnet associations
-

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags

Routes (1)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table

aws [Search] [Alt+S] United States (N. Virginia) Dhanam M (0517-4104-1226) Dhanam M

VPC > Route tables

Route table rtb-0900a7927086951c6 | private was created successfully. Last updated less than a minute ago

Route tables (4) info

Find route tables by attribute or tag

☐	Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☐	-	rtb-091c12286116cdf02	-	-	Yes	vpc-0be33b50250021280
☐	-	rtb-0364258cb337d9ef6	-	-	Yes	vpc-09a9894de98607334 myv.
☐	public	rtb-0d50ff40d2f8e9a42	-	-	No	vpc-09a9894de98607334 myv.
☐	private	rtb-0900a7927086951c6	-	-	No	vpc-09a9894de98607334 myv.

Then we have to create an internet gateway. After creation we have to attach to the VPC.

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VPC > Internet gateways

VPC dashboard < AWS Global View Filter by VPC

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Internet gateways (1/2) Info

Find internet gateways by attribute or tag

	Name	Internet gateway ID	State	VPC ID
☐	-	igw-00a777295cd7d64d9	Attached	vpc-0be33b50250021280
☒	myinternetgateway	igw-09a4ed4a4fa4b15cd	Detached	-

Actions: View details, Attach to VPC, Detach from VPC, Manage tags, Delete internet gateway

igw-09a4ed4a4fa4b15cd / myinternetgateway

Details Tags

Details

Internet gateway ID	State	VPC ID	Owner
igw-09a4ed4a4fa4b15cd	Detached	-	051741041276

Attach to VPC (igw-09a4ed4a4fa4b15cd) Info

VPC
Attach an internet gateway to a VPC to enable the VPC to communicate with the internet. Specify the VPC to attach below.

Available VPCs
Attach the internet gateway to this VPC.

Q vpc-09a9894de98607334 X

► AWS Command Line Interface command

Cancel Attach internet gateway

Internet gateway igw-09a4ed4a4fa4b15cd successfully attached to vpc-09a9894de98607334

Internet gateways (2) Info

Find internet gateways by attribute or tag

	Name	Internet gateway ID	State	VPC ID	Owner
☐	-	igw-00a777295cd7d64d9	Attached	vpc-0be33b50250021280	051741041276
☐	myinternetgateway	igw-09a4ed4a4fa4b15cd	Attached	vpc-09a9894de98607334 myvpc	051741041276

Creation of NAT Gateway.

VPC > NAT gateways > Create NAT gateway

Name - optional
Create a tag with a key of 'Name' and a value that you specify.
mynat
The name can be up to 256 characters long.

Availability mode Info
Choose whether to deploy across all zones in the region or restrict to a single availability zone.

☒ **Regional - new**
Scales automatically across all regional AZs, simplifying management for multi AZ deployments.

☐ **Zonal**
Provides granular control within a specific availability zone, adhering to subnet level settings.

VPC
Select a VPC in which to create the regional NAT gateway.
vpc-09a9894de98607334 (myvpc)

Connectivity type
Select a connectivity type for the NAT gateway.
☒ **Public**
☐ Private

Method of Elastic IP (EIP) allocation Info
Choose how IP addresses are associated with NAT gateways.

☒ **Automatic**
AWS automatically manages EIPs and AZ coverage for NAT gateways. This ensures easy scaling-- adding AZs automatically allocates EIPs, simplifying management.

☐ **Manual**
Manually assigns specific IP addresses for compliance or whitelisting. Note: Requires manual scaling to new AZs as workloads expand.

Tags
A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

Key **Value - optional**

Q Name X Q mynat X Remove

VPC > NAT gateways

VPC dashboard <
AWS Global View
Filter by VPC

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NAT gateway nat-16ee0c031115af76e | mynat was created successfully.

NAT gateways (1) Info Actions Create NAT gateway

Find NAT gateways by attribute or tag

Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID	P
mynat	nat-16ee0c031115af76e	Public	Pending	-	Regional	-	-

Then we have to create two security groups for the public and private instance.

For public instance inbound rules are.

Inbound rules Info

Type Info **Protocol** Info **Port range** Info **Source** Info **Description - optional** Info

SSH TCP 22 Anywh... 0.0.0.0/0 Delete

All ICMP - IPv4 ICMP All Anywh... 0.0.0.0/0 Delete

Add rule

For private instance inbound rules are.

Inbound rules Info

Type Info **Protocol** Info **Port range** Info **Source** Info **Description - optional** Info

All traffic All All Custom sg-0bb7a979c97fa9142 sg-0bb7a979c97fa9142 Delete

Add rule

Association of subnets to Route Table.

For private Route Table.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☐	public-subnet	subnet-04059115aa5508c21	10.0.1.0/24	-	rtb-0d50ff40d2f8e9a42 / public
☒	private-subnet	subnet-07428dba21a0407	10.0.2.0/24	-	Main (rtb-0364258cb337d9ef6)

Selected subnets

subnet-07428dba21a0407 / private-subnet

[Cancel](#) [Save associations](#)

Then Route also

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	NAT Gateway	-	No	CreateRoute

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

For public Route Table.

Edit subnet associations

Change which subnets are associated with this route table.

Available subnets (1/2)

Filter subnet associations

☒	Name	Subnet ID	IPv4 CIDR	IPv6 CIDR	Route table ID
☒	public-subnet	subnet-04059115aa5508c21	10.0.1.0/24	-	Main (rtb-0364258cb337d9ef6)
☐	private-subnet	subnet-07428dba21a0407	10.0.2.0/24	-	Main (rtb-0364258cb337d9ef6)

Selected subnets

subnet-04059115aa5508c21 / public-subnet

[Cancel](#) [Save associations](#)

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

[Add route](#)

[Cancel](#) [Preview](#) [Save changes](#)

Then we have to Launch the public and private instance.

For public instance.

EC2 > Instances > Launch an instance

Network settings

VPC - required | Info
vpc-09a9894de98607334 (myvpc) 10.0.0.0/16

Subnet | Info
subnet-04059115aa5508c21 public-subnet
VPC: vpc-09a9894de98607334 Owner: 051741041276 Availability Zone: us-east-1a (use1-az1) Create new subnet
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP | Info
Enable

Firewall (security groups) | Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups | Info
Select security groups
public-sg sg-0bb7a979c97fa9142 X
VPC: vpc-09a9894de98607334
Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

Configure storage

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Summary

Number of instances | Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...read more
ami-02dfbd4ff395f2a1b

Virtual server type (instance type)
t3.micro

Firewall (security group)
public-sg

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

For private instance.

Network settings

VPC - required | Info
vpc-09a9894de98607334 (myvpc) 10.0.0.0/16

Subnet | Info
subnet-07428dbaac21a0407 private-subnet
VPC: vpc-09a9894de98607334 Owner: 051741041276 Availability Zone: us-east-1a (use1-az1) Create new subnet
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.2.0/24

Auto-assign public IP | Info
Disable

Firewall (security groups) | Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
☐ Create security group ☒ Select existing security group

Common security groups | Info
Select security groups
private-sg sg-095be24b3a2fac96f X
VPC: vpc-09a9894de98607334
Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

Configure storage

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Summary

Number of instances | Info
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...read more
ami-02dfbd4ff395f2a1b

Virtual server type (instance type)
t3.micro

Firewall (security group)
private-sg

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

Then I have checked the network connection using the ping command.

```

[ec2-user@ip-10-0-1-14 ~]$ ping 3.218.156.195
ping 3.218.156.195 (3.218.156.195) 56(84) bytes of data:
64 bytes from 3.218.156.195: icmp_seq=1 ttl=126 time=0.229 ms
64 bytes from 3.218.156.195: icmp_seq=2 ttl=126 time=0.223 ms
64 bytes from 3.218.156.195: icmp_seq=3 ttl=126 time=0.213 ms
64 bytes from 3.218.156.195: icmp_seq=4 ttl=126 time=0.213 ms
64 bytes from 3.218.156.195: icmp_seq=5 ttl=126 time=0.229 ms
64 bytes from 3.218.156.195: icmp_seq=6 ttl=126 time=0.231 ms
64 bytes from 3.218.156.195: icmp_seq=7 ttl=126 time=0.254 ms
64 bytes from 3.218.156.195: icmp_seq=8 ttl=126 time=0.254 ms
64 bytes from 3.218.156.195: icmp_seq=9 ttl=126 time=0.226 ms
^C
--- 3.218.156.195 ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8325ms
rtt min/avg/max/mdev = 0.213/0.230/0.254/0.014 ms
[ec2-user@ip-10-0-1-14 ~]$

```

Then I have connected to the private instance through the public instance with the help of the private ip of the private instance.

PublicIPs: 3.218.156.195 PrivateIPs: 10.0.1.14

```

[tc2-user@ip-10-0-2-222 ~]$ ping 10.0.2.222
ping 10.0.2.222: 64 bytes of data:
64 bytes from 10.0.2.222: icmp_seq=1 ttl=127 time=0.027 ms
64 bytes from 10.0.2.222: icmp_seq=2 ttl=127 time=0.031 ms
64 bytes from 10.0.2.222: icmp_seq=3 ttl=127 time=0.031 ms
64 bytes from 10.0.2.222: icmp_seq=4 ttl=127 time=0.040 ms
64 bytes from 10.0.2.222: icmp_seq=5 ttl=127 time=0.038 ms
^C
10.0.2.222 ping statistics:
  packet sent: 5, received: 5, % packet loss, time 4130ms
  rtt min/avg/max/mdev = 0.027/0.033/0.040/0.004 ms
[tc2-user@ip-10-0-2-222 ~]$

```